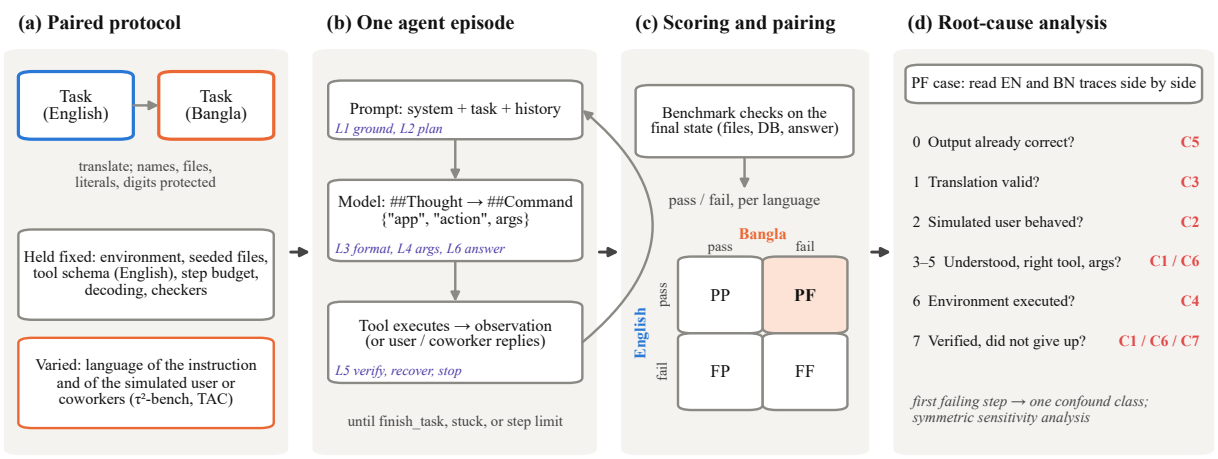

(a) Paired protocol
Task (English)
Task (Bangla)
translate; names, files, literals, digits protected
Held fixed: environment, seeded files, tool schema (English), step budget, decoding, checkers
Varied: language of the instruction and of the simulated user or coworkers (τ²-bench, TAC)
(b) One agent episode
Prompt: system + task + history
L1 ground, L2 plan
Model: ##Thought → ##Command {"app", "action", args}
L3 format, L4 args, L6 answer
Tool executes → observation (or user / coworker replies)
L5 verify, recover, stop
until finish_task, stuck, or step limit
(c) Scoring and pairing
Benchmark checks on the final state (files, DB, answer)
pass / fail, per language
Bangla
pass
fail
English
pass
fail
PP
PF
FP
FF
(d) Root-cause analysis
PF case: read EN and BN traces side by side
0 Output already correct? C5
1 Translation valid? C3
2 Simulated user behaved? C2
3–5 Understood, right tool, args? C1 / C6
6 Environment executed? C4
7 Verified, did not give up? C1 / C6 / C7
first failing step → one confound class; symmetric sensitivity analysis